Stigma Free Lunch: School Meals and Student Discipline

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Abstract

We investigate the effect of the Community Eligibility Provision (CEP) of the National School Lunch Program on student suspension rates. CEP allows high-poverty schools to offer free meals to all students regardless of their family's household income. We conceptualize CEP as a strategy to alleviate well-documented stigma associated with school meals. As such, we hypothesize that CEP implementation will reduce the incidence of suspensions, particularly for students from low-income backgrounds and minoritized racial and ethnic groups. We construct a unique panel dataset that links educational records for K-12 students enrolled in Oregon public schools between 2010 and 2017 with rich administrative data records describing their families' household income and social safety net program participation. Difference-in-difference analyses indicate that CEP implementation has a measurable protective effect on suspension rates in participating schools. These effects are pronounced for students from low-income families and Hispanic students.

Keyword*:* Community Eligibility Program; School Discipline; School Lunch.

JEL Classification: I21, I24, I28, I32

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"I learned I was poor on the new lunch lady's first day at work."

- Psychologist Kenneth Payne, The Broken Ladder: How Inequality Affects the Way We Think, Live, and Die (2017, p. 11)

Elementary and secondary schools distribute free or subsidized meals to some 30 million students across the United States on the average school day via the National School Lunch Program (NSLP), at a cost of approximately $14 billion annually (USDA 2022). These meals represent one of the largest child nutrition and development programs in the contemporary U.S. and provide a crucial nutritional supplement to young people across the country who are at risk of food insecurity (Gordon et al. 2007).

Schools collect data on students' family income and use these data to determine which students are eligible to receive free or subsidized meals. In doing so, schools enact social categories that are closely linked to student socioeconomic background. Schools reinforce these income-based categories via daily rituals that demarcate poor and non-poor students. For example, schools may ask students from relatively low-income backgrounds to line up for school meals and produce a lunch ticket to pay for their meals while more affluent students sit to eat their bagged lunches (Best 2017; Poppendieck 2011).

In this paper, we investigate the consequences of these income-based school lunch categories by estimating the effects of a policy that eliminates them: The Community Eligibility Provision (CEP) of NSLP. When schools implement CEP, they offer free meals to all students regardless of their family's household income, suspending processes that distinguish between students who receive free or subsidized meals and those who must pay full price. Eligible schools and school districts across the United States have implemented CEP in a staggered fashion since the program's launch in the 2014-15 school year, with a smaller number of states starting earlier under the pilot program. At the start of the 2019-20 school year, nearly 70 percent

of eligible schools –approximately 1 in 3 public K-12 schools – participated in CEP (FRAC 2020).

Our work, which developed in the context of a broader project assessing the use of free and reduced-price lunch data as a proxy for student economic disadvantage (Domina et al. 2018) and responds to calls for increased attention to the foundational role that school meals play in the social organization of the school and the development of youth identities (Best 2017, Weaver-Hightower 2011), conceptualizes CEP as an intervention that alleviates a source of socioeconomic stigma in participating schools. Alleviating the stigma associated with free and reduced-price lunch likely has diverse consequences in the social order of the school, including encouraging more students to participate in school meals and reducing the salience of social class in teacher expectations and peer dynamics. Drawing upon prior scholarship that positions stigma at the heart of a vicious cycle that drives exclusionary discipline (Okonofua et al. 2016) – we focus attention on the effects of CEP on student suspensions. Prior research suggests that the implementation of CEP and other universal school meal programs boost student achievement (Gordanier et al. 2020, Kitchen et al. 2013; Ruffini 2021, Schwartz & Rothbart 2020) and attendance (Bartfield, Berger, & Men 2020). While limited evidence regarding the program's effect on suspensions is currently available (Gordon & Ruffini 2020; Kho 2018), we hypothesize that CEP reduces the incidence of suspensions in treated schools and that this negative effect increases over time as the stigma associated with free lunch fades and schools implement new universal free lunch routines. We further hypothesize that the program's benefits will be particularly pronounced for students from groups that are most at risk of group-based stigma in school, including students from socioeconomically disadvantaged families, and students from minoritized racial and ethnic groups.

Our analyses take advantage of a unique dataset consisting of administrative data describing all students enrolled in Oregon public schools between 2010 and 2017 merged with household-level data from the U.S. Census Bureau describing Internal Revenue Service (IRS)-reported earnings and Supplemental Nutrition Assistance Program (SNAP) participation. Focusing analyses on 567,000 elementary, middle, and high school students enrolled in schools that were eligible to participate in CEP, we execute a difference-in-differences study design, comparing changes in suspension rates for students enrolled in schools that implement CEP with contemporaneous suspension rate changes for students in schools that do not implement CEP. Our design, which takes advantage of CEP's staggered implementation across eligible schools, generates plausibly causal estimates of program treatment by assuming suspension trends for schools that implement CEP would move in parallel to trends for schools that do not, if not for the implementation of CEP. Although CEP rolled out in a period of rapid decline in the use of suspensions in Oregon schools, our analyses indicate that this identifying assumption holds.

We find that CEP implementation led to a further decline in suspensions for students in treated schools, with statistically significant effect size estimates equivalent to an approximately 15 percent decline in suspension rates in the third year of CEP implementation. Our unique student-level data linkages – and, in particular, IRS and SNAP data that characterize family economic resources – allow us to investigate the distinctive effects of CEP implementation on students from low-income families, during a period where the NSLP indicator becomes unreliable as a proxy for family income. We find that CEP has a particularly pronounced protective effect for low-income students, as well as Hispanic students (the state's largest minoritized student group). These subgroup findings, in particular, represent a new contribution to our understanding of which students are most affected by school lunch stigma and its removal

via CEP. They also highlight the critical importance of reliable measures of family economic circumstances in education research.

Community Eligibility Provision and universal school lunch

Meal programs in U.S. schools

Founded in 1946 and expanded considerably in the late 1960s and early 1970s, the National School Lunch Program has often been primarily conceptualized as an anti-poverty program that targets free and low-price meals to students at risk of food insecurity (Levine 2008). USDA operates NSLP, and its regulations require schools to provide free or reduced-price lunch application to all students at the beginning of the school year and make applications available upon request at any time. These applications ask students to demonstrate their eligibility for free or reduced-price lunch by reporting their household's total income from all sources in the prior month. Under the policy, students who demonstrate income-eligibility remain enrolled for the entirety of a school year. Students qualify for free lunches if their families report household incomes less than 130 percent of poverty and reduced-price lunches if they report income between 130 and 185 percent of poverty. Students who do not apply for the program or whose families report income greater than 185 percent of poverty must pay for school lunches.

Several policy initiatives implemented since the 1980s have attempted to streamline the process of enrolling in free or reduced-price lunch, allowing students whose families participate in SNAP, Temporary Assistance for Needy Families (TANF), and other means-tested federal programs as well as homeless, migrant, and foster children to enroll without submitting an application via a procedure known as "direct certification." Other relatively recent policy

initiatives allow participating schools to automatically re-enroll students who had previously demonstrated income eligibility without completing an annual application.

Authorized as part of the 2010 Healthy, Hunger-free Kids Act of 2010, CEP dramatically expanded and accelerated the policy movement to streamline and expand access to free school meals. Schools, groups of schools, or districts in which in which 40 percent or more of students directly certify for free lunch by participating in SNAP or other means-tested federal programs can opt into CEP. Once they do so, participating schools offer free lunch and breakfast to all students, regardless of their demonstrated need, receiving funding under a National School Lunch Program reimbursement formula. The formula, however, makes the program less appealing for schools and districts that are just barely eligible – these schools run the risk of not covering their meal expenses through NSLP reimbursement alone, and may have to dip into other revenue sources to cover universal breakfast and lunch expenses. These incentives leave plenty of room for eligible schools to opt in or out based on the financial and political support for the program locally.

Existing evidence on the effects of universal school meals

Recent evaluations of CEP implementation in U.S. schools as well as studies of universal meal provision in other policy contexts indicate that universal meal provision boosts student participation in school meals (Schwartz & Rothbart 2019; Ruffini 2021; Turner, Guthrie & Ralston 2019; Pokorney, Chandran & Long 2019; Rothbart, Schwartz, & Gutierrez 2020; Gordanier et al. 2020; Kitchen et al. 2013). While it is difficult to directly compare the magnitude of changes in meal participation across settings, the results typically suggest that students in schools with universal lunch provision are 5-10 percentage points more likely to participate in school lunch than they were in means-tested lunch provision. This literature

provides some evidence to suggest that increases in meal participation are particularly pronounced for the relatively economically advantaged students who would have previously had to pay for meals. However, these studies find positive effects in meal receipt for lower-income free or reduced-price eligible students as well.

This growing research literature further indicates that universal meal provision positively affects students' academic and behavioral outcomes. Several studies indicate that universal meal provision modestly but significantly boosts student math achievement, with average effect size estimates ranging from 0.04 to 0.06 standard deviations (Bartfield, Berger, & Men 2020; Gordanier et al. 2020; Kitchen et al 2013; Ruffini 2021). While there is some evidence to suggest that universal meals have similar positive effects on reading achievement (Schwartz & Rothbart 2019), other studies show no effects on this achievement outcome (Gordanier et al. 2020, Ruffini 2021). Further, the existing literature indicates that universal meal provision improves elementary school student attendance (Gordanier et al. 2020; Barfield, Berger & Mann 2020), though it is unclear whether these positive effects generalize to secondary school settings.

Relatively little evidence is currently available regarding the effects of universal meals on student disciplinary outcomes. Using data from New York City, Gutierrez (2020) demonstrates that school-level universal free meals program adoption drives notable improvements in student perceptions of school climate, including reported declines in bullying and fighting in schools. Gordon & Ruffini (2020) take advantage of the fact that eight states participated in a CEP pilot program starting in 2012 (before the program extended nationwide in 2015) to estimate the effects of attending a CEP-eligible school on suspension rates and other disciplinary outcomes. Gordon & Ruffini's analyses use data from four waves of the Department of Education's Civil Rights Data collection, which reports the number of students suspended from more than 90

percent of U.S. public schools biennially since 2010. These analyses suggest that suspension rates for elementary and middle school students in CEP-eligible schools fall modestly when CEP becomes available to schools in their states. While informative, these estimates likely understate the effect of school-level CEP participation, since only half of eligible schools had opted into the program by 2017. More directly relevant are findings from Kho's (2018) unpublished dissertation using data from Tennessee. Using a difference-in-differences approach, Kho finds that student suspension rates remain unchanged during a school's first year of CEP participation but decline by 1.5 percentage points in their school's 2nd year in the program and 2.3 percentage points in their school's 3rd year in the program.

A sociological account of the effects of universalizing school meals

Many existing evaluations of CEP and other universal meal programs invoke the concept of stigma to explain the program's effects. Goffman (1963, p. 3) describes stigma as an "attribute that makes [its holder] different from others in the category of persons available for him to be, and of a less desirable kind – in the extreme, a person who is quite thoroughly bad, or dangerous, or weak. He is thus reduced in our minds from a whole and usual person to a tainted, discounted one." Link & Phelan (2001, p, 367) elaborate, writing that "we apply the term stigma when elements of labeling, stereotyping, separation, status loss, and discrimination co-occur in a power situation that allows the components of stigma to unfold." We argue that means-testing activates stigma by associating school meals with poverty.

Students who come of age in schools that provide free or reduced-price meals to income-eligible students often use terms such as "disgusting" and "welfare food" to describe school meals (Poppendieck 2010). Participation in means-tested free- or reduced-price school lunch programs marks students as disadvantaged in the eyes of their teachers and peers, who consider

packed lunches from home as a mark of status and affluence (Weaver-Hightower 2011; Best 2017). The processes through which schools distribute means-tested meals may contribute to school meal stigma, by shaping teachers' impressions of students who eat school meals and creating both physical and social distance between meal recipients and their school peers. This stigma may lead many students – including the income-eligible and their relatively affluent peers – to opt out of school-provided meals (Mirtcheva & Powell 2009). Further, it likely raises the salience of student social class and other ascriptive student characteristics associated with social class in school status hierarchies.

School meals and teacher perceptions

One way the means-tested allocation of school meals create stigma is by shaping teachers' expectations for student behavior and educational achievement. Generations of scholarship in the sociology of education draws attention to the ways student social class structures the relationship between teachers and administrators, who typically hail from middle class backgrounds, and their socioeconomically diverse students. Rist's (1970) documents how information that teachers gather from social workers, fellow teachers, and interactions with parents about their students' family backgrounds shapes teacher expectations of student behavior and ability even before teachers begin to interact substantially with students. In schools in which school meal distribution is means-tested, teachers distribute applications for free- or reduced-price lunch applications to students and their families, encouraging families to apply, and collecting and submitting applications once they have been completed, and then organizing students into lines at meal time. As such, the means-tested allocation of school lunches likely provides teachers with information to infer students' socioeconomic status. These inferences can have long-lasting consequences: A rich body of sociological research demonstrates that can help

to explain why teachers underestimate poor students' academic ability (Ready & Wright 2011), motivation (Auwarter & Aruguete 2008; Jussim et al. 1987), and behavior (Dunkake & Schuchart 2015, Calarco 2018).

School meals and student social dynamics

Furthermore, since school meals represent one of the few periods in the school days in which young people have relative autonomy, students' lunchtime socializing often defines their social experiences in the school, shaping both the way students see themselves and the way they are seen by others in the school environment (Milner 2013, Best 2017; Calarco 2018). As students sort into school lunch groupings, they often develop peer subcultures, distinguished by distinct norms of dress and self-presentation, behavior, and status ranking in the school's social hierarchy. Reflecting on the school lunchroom's role in his high school's social life, one student recounts: "[There was] a great deal of concern as to what the lunch arrangements would be. As freshmen, people scurried around the lunch room, searching desperately for friends to sit with, so that they would not have to suffer the ultimate embarrassment of eating alone or sitting at an uncool table." Another adds: "I always felt that my level of coolness was being judged by the respective status of each individual I was with." (Milner 2013, p. 61).

The mechanisms of means-tested lunch distribution may increase the salience of social class at mealtime and throughout the school day. To receive school meals, students must line up and collect their food on a distinctive school meal tray before they sit and eat. In many schools they must further disclose their meal status at the point of payment. Students who opt out of school meal programs, meanwhile, can often proceed straight to their meals. Furthermore, as a growing number of schools begin to offer unsubsidized a la carte meal options, students who can afford to do so may skip the school meal line and pay by cash for more desirable snack foods

(Best 2017; Poppendieck 2011). In each of these cases, the processes of means-tested school meal distribution sort and spatially segregate students, influencing how students socialize in the lunchroom and contributes to the association between social class and student social networks and identities (Best 2017).

Meal stigma and school discipline

Meal stigma likely has a wide range of negative consequences. In some cases, stigma may lead students to skip school meals, going hungry rather than risking the stigma associated with poverty. In other cases, free meal participation may mark students (in their own eyes, in the eyes of their peers, and in their teacher's eyes) as poor. By allowing participating schools to offer free meals to all students, the CEP renders the means-tested meal distribution mechanisms unnecessary, arguably mitigating meal stigma and its consequences for students.

Doing so, we hypothesize, has downstream consequences for the disciplinary climate in participating schools. We imagine two mechanisms through which these consequences might emerge:

First, universal meal provision may improve student behavior by improving student nutrition. Prior research clearly demonstrates that CEP and other universal meal provision programs boost student participation in school meal programs (Schwartz & Rothbart 2019; Ruffini 2021; Turner, Guthrie & Ralston 2019; Pokorney, Chandran & Long 2019; Rothbart, Schwartz, & Gutierrez 2020; Gutierrez 2021; Gordanier et al. 2020; Kitchen et al. 2013). A separate literature indicates that youth hunger and malnutrition are associated with externalizing behaviors (Howard 2011; Kimbro & Denney 2015) and nutritional supports protect at-risk youth from suspensions and other forms of school discipline (Gennetian et al. 2016).

Second, if universal meal provision reduces stigma and stereotyping associated with poverty in schools, the CEP may interrupt vicious cycles that contribute to high rates of exclusionary discipline, particularly among students who participate in free and reduced-price lunch. Existing scholarship on disciplinary disparities focuses extensive attention on pronounced and persistent racial disparities in exposure to school discipline (Anderson & Ritter 2017; Barrett, McEachin, Mills & Valant 2021; Gregory, Skiba, & Noguera 2010; Lindsay & Hart 2017; Owens & McLanahan 2020; Freidus 2020). This research literature focuses on both educator biases and student disengagement. Teachers, primed by racist stereotypes, subject Black and Hispanic students to sharp behavioral scrutiny at school and issue disproportionately draconian punishments for even minor infractions (Gilliam et al. 2016; Okonofua & Eberhardt 2015). In response, students of color come to believe that school is not for them and may behave accordingly (Borman et al 2021; Goyer et al. 2019; Kupchik, 2010). Although disciplinary disparities based on social class and school meal participation are less well-studied than racial disparities, students who are enrolled in school free or reduced-price lunch program are suspended from school at nearly twice the rate of their more affluent peers (Anderson & Ritter 2017, Barrett et al. 2019, Owens & McLanahan 2019). Traditional means-tested school meal programs may contribute to these disparities by marking meal participants as poor, subjecting lunch recipients to particular scrutiny as they activate teachers' and peers' broader class-based biases, potentially on top of, or in exacerbation of, race-based biases.

The nutrition and stigma mechanisms are entwined, because stigma is what prevents some eligible students from eating free lunch. The fact that universal free meal provision boosts uptake among previously ineligible students as well as low-income students suggests that the

association of low socioeconomic status and free lunch fades alongside its nutritional reach. We therefore hypothesize that:

1. CEP reduces the incidence of suspensions in treated schools.
2. The effect of CEP on suspensions increases over time as schools implement new universal free lunch routines and the stigma associated with free lunch fades.
3. CEP has a particularly pronounced protective effect on suspensions for low-income students.
4. CEP has a pronounced protective effect on suspensions for Hispanic students, the largest minoritized racial/ethnic group among Oregon public school students.

Data

To test these hypotheses, we link student-level educational administrative data from the Oregon Department of Education (ODE) with confidential microdata stored at the Census Bureau including SNAP records from the state of Oregon and income tax records from the IRS. The ODE data provide annual records on all students enrolled in Oregon public K-12 schools between 2009-10 and 2016-17. Students appear in SNAP when they are listed as members of households in SNAP records; they appear in IRS records when they are claimed as dependents on IRS tax returns.

Merging across these data sources is possible using Census's Protected Identification Keys (PIKs), which are assigned to the three administrative data sources using the Person Identification Validation System.[2] As noted in Table 1, 96 percent of K-12 students in the ODE

[2] This is a probabilistic matching approaching conducted by the Census Bureau that leverages personally identifiable information in the raw administrative data to assign an anonymous unique identifier known as a PIK. After the assignment of PIKs, personally identifiable information is removed to protect confidentiality and these anonymized files can be accessed by approved researchers. For more information on the Person Identification Validation System, see Wagner and Layne (2014).

data are assigned PIKs in this process. We exclude students without PIKs from our analyses since we need to be able to link students to the other data sources.

Appendix A provides details on the variables we construct using the linked data. One variable of particular importance is the low income indicator, which leverages both SNAP and IRS data to identify students who likely live in households with income below 1.85 times the federal poverty line. In addition to restricting our sample to students who have PIKs, we further require a non-missing value of the low income variable, since this variable is central to our heterogeneity analyses. As shown in table 1, 6% of Oregon students do not have a low income value, either because they do not have a PIK or because they do not appear on an IRS Form 1040 record within the three years prior to the reference year and they do not receive SNAP at any point during the reference year. Our investigation of students missing the low income flag did not support the assumption that all of these students are low income, thus, in dropping them, we proceed with the implicit assumption that they are missing at random. After restricting based on the availability of PIK and income information, the data is reduced to 4.2 million student-year observations.

To create our analytic sample, we additionally exclude students enrolled in schools that are not eligible to participate in CEP, as well as a small number of students enrolled in extremely small schools (schools with fewer than 20 students), specialty schools managed directly by their district or the Oregon Department of Education, and schools that either opened or closed at any point during the study period. Schools are eligible for CEP if 40 percent or more of enrolled students directly certified for free school meals. Data on this measure, which the USDA refers to as the Identified Student Percentage (ISP), are available for Oregon between 2014-15 and 2016-17 from the Food Research and Action Center (FRAC) and the Center on Budget and Policy

Priorities. However, since these data may be incomplete, we additionally use data on school SNAP rates, because SNAP participation is the main characteristic used to calculate ISP. Schools are marked as eligible for CEP if SNAP percentages are greater than 40 percent in the prior school year, since the prior year's rate determines current-year CEP eligibility. Finally, because districts can group schools into CEP-eligible clusters based on the group ISP rate, we marked eligible all schools in districts where the entire district has a SNAP percentage of at least 40 percent. This includes schools that may not be individually eligible for CEP, but are eligible as part of a full district. These sample restrictions exclude roughly one-half of the full data, for an analytic sample of 567,000 students and 2,104,000 student-year observations.

Table 1 highlights the ways that students in CEP eligible schools differ from the average Oregon student. Since CEP is only available to schools and districts in which 40 percent or more of students directly certify for free lunch by enrolling in SNAP or other means-tested social service programs, our analytic sample includes a higher proportion of low-income students, English language learners, and Hispanic students than the state overall. The analytic sample also has somewhat higher average suspension rates. Elementary schools are disproportionately likely to be eligible for CEP and, accordingly, CEP schools have lower enrollment than the state average. More notably, CEP-eligible schools are more likely to be enrolled in Provision 2, a CEP predecessor that also provided free lunch to all students within a school, but was less widely adopted compared to CEP. Schools that enroll in CEP must exit the Provision 2 program, so we designate Provision 2 involvement as enrollment prior to 2014-15.

Empirical methodology

We employ a difference-in-differences (DiD) approach to estimate the effects of CEP implementation on suspension rates in CEP-eligible schools, with particular attention to heterogeneous treatment effects across the student subgroups most likely to confront stigma. In recent years, the econometrics literature has paid substantial attention to the complexity of identifying treatment effects using a DiD model when entry into treatment is staggered across time (Goodman-Bacon 2021). In particular, the canonical model with a single treatment effect is inadequate when treatment effects build over time, as we hypothesize they do here.

We have three treatment cohorts – schools that begin CEP in 2014-15 (the 2015 cohort), 2015-16 (the 2016 cohort), or 2016-17 (the 2017 cohort) – and, thus, must employ a DiD method that works for staggered entry with time-varying treatment effects. We take the approach outlined by Wooldridge (2021) which establishes clear control groups and allows treatment effects to vary by cohort and time period of treatment. Of course, estimating different treatment effects for each cohort and time period yields a large number of estimates, so we show we can reasonably assume common effects across cohorts in each treatment period to arrive at our central model. Ultimately, we estimate the treatment effect of CEP in the first, second, and third years after schools adopt the program, both overall and for a number of subgroups of interest.

Estimating assumptions

Two key assumptions must hold for a DiD model to generate an unbiased causal treatment effect. First, we assume that schools did not anticipate treatment and, therefore adjust policies and practices in advance of implementation. This is a reasonable assumption in this setting since eligible schools and districts act in late spring to enroll in CEP for the subsequent school year. It is unlikely that students, teachers, or administrators change their behavior with respect to suspension because they anticipate universal free lunch provision the next year. It is

also unlikely that schools change their lunch practices prior to their official enrollment in CEP, since their reimbursements do not change until after enrollment. Second, we assume that treatment and control group outcomes would move in parallel if not for the treatment. We empirically investigate the parallel trends assumption below.

A further complication to our proposed DiD approach is the potential for selective exit in our treatment sample. Schools enrolling in CEP receive access to the program for four years, so none are forced out of the program during our period of study. Schools can, however, opt out of the program between years. We speculate that such a choice indicates that schools did not accurately anticipate the costs of participating in CEP relative to their prior meal costs before they chose to participate. Since exit occurs between school years, there is no exit during the first treatment period. Just one school exits from the 2015 cohort in the 2015-16 school year, while 20 (of approximately 300) schools exit the 2015 and 2016 cohorts in the 2016-17 school year. We opt to ignore this exit and, thus, treat exiting schools as though they remained in CEP. This potentially introduces a conservative bias[3] to our estimates – particularly those pertaining to the 2017 school year.

Estimating the effect of CEP implementation

Our central DiD model identifies average treatment-on-the-treated effects for the first, second, and third periods of CEP implementation. To support our assumption that treatment effects do not vary across cohorts (i.e., CEP has the same effect in the first year regardless of which year a school starts the program), we first estimate a disaggregated version of the central

[3] There is no theoretical reason to think that offering universal free lunch one year and then removing it the next leads to improved suspension outcomes relative to offering universal free lunch in both years. If schools stop providing universal free lunch, they likely lose both potential benefits of CEP: improved food access and stigma reduction. It remains possible, however, that schools find ways to maintain aspects of CEP they found beneficial as they transition back to the old reimbursement system.

model with cohort-by-period treatment effects.[4] This allows us to test whether treatment effects, in fact, vary across cohorts. We then extend the central model to identify heterogeneous treatment effects across student subgroups.

Our central model is as follows:

$$(1) \quad y_{itj} = \eta + \sum_{s=11}^{17} \theta_s fs_t + \sum_{p=1}^{3} \tau_p g_{j,p} + \gamma_j + \epsilon_{it}$$

The indicator $g_{j,p}$ flags the year of CEP participation for treated schools, where p ranges from 1-3. The corresponding coefficient, τ_p, represents the period-specific treatment effect.[5] Our second hypothesis holds that the value of τ_p should increase as p increases. Treatment effects are identified under our estimating assumptions because school fixed effects (γ_j) remove level differences in suspensions rates across schools, and academic year indicators fs_t adjust for time trends in suspension rates common to all CEP-eligible schools (note: index s ranges from 2011 through 2017, with 2010 excluded as a reference year).

Estimating variation in effects of CEP across subgroups

In an analysis central to our paper's contribution, we further estimate the treatment effects for different subgroups of students. For simplicity, consider subdividing students into two

[4] This model takes the following form:

$$y_{itj} = \eta + \sum_{s=11}^{17} \theta_s fs_t + \tau_{15,15} d_{i,15} f15_t + \tau_{15,16} d_{i,15} f16_t + \tau_{15,17} d_{i,15} f17_t + \tau_{16,16} d_{i,16} f16_t + \tau_{16,17} d_{i,16} f17_t + \tau_{17,17} d_{i,17} f17_t + \gamma_j + \epsilon_{it}$$

y_{it}, the dependent variable in this model, as in all models we estimate in this paper, is an indicator for whether student i is suspended in year t while enrolled in school j. Each treatment cohort receives its own treatment indicator, $d_{i,y}$, where y=15 if the school entered treatment in 2014-15 academic year, and so on. In addition, the model includes academic year indicators (fs_t where s ranges from 2011 through 2017, with 2010 excluded as a reference year), and the interaction of treatment indicators with academic year indicators *for academic years in which the cohort is participating in CEP*. The coefficients on these interaction terms ($\tau_{15,15}, \tau_{15,16}, \ldots, \tau_{17,17}$) capture the effects of CEP for each cohort and in each year the cohort is treated.

[5] Since our data ends in the 2016-17 school year, not all cohorts are reflected in each treatment period; while all three cohorts contribute to τ_1, the 2015 and 2016 cohorts contribute to τ_2, and only the 2015 cohort contributes to τ_1.

groups according to binary indicator B_i, which could denote having a low-income family, among other attributes.

$$(2) \quad y_{itj} = \eta + B_i + \sum_{s=11}^{17} \theta_s fs_t + \sum_{s=11}^{17} \pi_s fs_t B_i + \sum_{p=1}^{3} \tau_p g_{j,p} + \sum_{p=1}^{3} \rho_p g_{j,p} B_i + \gamma_j + \epsilon_{it}$$

In this formulation, ρ_1 would denote the difference in the treatment effect of CEP in its first period for students in group B compared to students not in group B. The treatment effect for group B students in the first period is $\tau_1 + \rho_1$, while the treatment effect for students not in group B remains τ_1.

We test our third hypothesis by estimating a version of model (3) that compares the effects of CEP implementation on students from low-income households and students from households that are not low income. We test our fourth hypothesis by estimating a version of model (3) that compares the effects of CEP implementation on American Indian/Alaska Native, Asian/Pacific Islander, Black, Hispanic, Multiracial, and White students. We additionally estimate a version that compare the effects of CEP implementation on students in grades K-5 and students in grades 6-12 and income-by-race subgroups. Because we are interested in how the net treatment effects vary across subgroups, rather than present the lengthy regression results implied by Model (3), we estimate linear predictions of treatment effects for different subgroups across the three treatment periods. For example, when we estimate the regression where B_i is an indicator for low income, we report the first period treatment effect for students who are not low

income ($\widehat{\rho_1}$), and contrast it to the first period treatment effect for students who are low income ($\widehat{\rho_1} + \widehat{\tau_1}$).[6]

Results

Patterns in CEP implementation

Figure 1 maps schools that implemented CEP across Oregon from 2014-15 to 2016-17 in black diamonds. The vast majority of participating schools took up CEP in 2015 (242), while 63 joined in 2016, and 11 joined in 2017. White circles indicate schools that did not participate in CEP, either because they were not eligible or because they opted not to take-up the program. The figure illustrates that CEP schools are distributed widely across the state of Oregon. The second map included in Figure 1 provides a close up view of CEP participation in the Greater Portland region. Most larger school districts in Oregon such as the Portland and Salem-Keizer school districts include both CEP-participating and non-participating schools. Approximately 40 school districts throughout Oregon include a mixture of participating and non-participating schools. In 22 districts, all schools participated in CEP.

[FIGURE 1 AROUND HERE]

As noted previously in Table 1, CEP-eligible Oregon public schools enroll more low-income students and more students of color than CEP-ineligible schools. This follows from the fact that federal policy only allows schools and districts with high proportions of economically disadvantaged students to participate in CEP. This pattern, however, also extends to program take-up among eligible schools, as shown in Table 2, which compares the demographic profile of

[6] We note that the inclusion of these subgroups changes the parallel trends assumption: For the results of Model (3) to yield unbiased estimates of the effect of CEP adoption by subgroup, we must assume that *subgroup* outcomes across treatment/control groups would move in parallel if not for CEP.

schools that implemented CEP during our study period (our treatment schools) with that of schools that were eligible to enroll in the program, but did not do so (our control schools). On average, treatment schools enroll a higher proportion of low-income students and students of color than control schools. For example, in 2014, treatment schools are 51 percent White and 34 percent Hispanic compared to control schools which are 66 percent White and 17 percent Hispanic. 49 percent of students in treatment schools are enrolled in SNAP in March 2014 compared to 37 percent of students in control schools. Students in treatment schools come from families with annual incomes $10,000 less, on average, than students in control schools. The CEP's funding formula, which reimburses participating schools based on the proportion of students who participate in SNAP and other means-tested social safety net programs, likely accounts for these patterns. Since this formula may make the program expensive to operate in economically diverse eligible schools, it is not surprising that the schools that enroll in CEP are among the lowest income schools in the state.

[TABLE 2 AROUND HERE]

Patterns of change in treatment and control schools: Testing the parallel trends assumption

Causal inferences based on difference-in-differences analysis relies on the assumption that trends between treatment and control groups would move in parallel if not for the treatment. It allows, however, that groups have different baseline outcomes. We provide a formal test of the parallel trends assumption, but first we summarize some descriptive evidence that our treatment and control schools, while different in average demographics, are subject to similar forces.

Table 2 describes covariate trends over time for students attending treatment ("ever in CEP") schools and control ("never in CEP") schools. We find little evidence of divergence in covariate trends between treatment and control groups. While these comparisons do not directly

test the parallel trends assumption, their results support the logic that these groups of schools were exposed to similar pressures during the pre-treatment period and that – by extension – divergences in outcomes that emerge during the post-treatment period are likely attributable to the effects of CEP enrollment. The percentage of Hispanic students rises by four percentage points in both treated and untreated schools over the 4-year pre-treatment time frame. Schools in both groups also see small parallel increases in the proportion of students who speak a language other than English in the home between 2010 and 2014. Similarly, although the mean IRS-reported income for students in both sets of schools increases by 5-6 percent between 2010-2014, the percent of students enrolled in SNAP increases by four percentage points in each group during this pre-treatment period.[7]

Figure 2 reports trends in suspension rates, our outcome variable. These unconditional trends provide additional evidence to support the parallel trend assumption. In 2010, the percentage of students ever suspended was 1.3 percentage points higher for students attending treatment schools compared to students attending schools that never participated in CEP. While suspension rates remain largely unchanged between 2010 and 2011 for schools in the control group, we observe substantial declines in suspension rates for both treated and control groups between 2011 and 2014. Suspension rates in both groups declined by approximately 1.5 percentage points. This rate of decline corresponds to a broad national trend in school discipline (Hirschfield 2018. In 2015, however, the first year of CEP, the trends begin to diverge. Students in treatment schools see a sustained decline in suspension rates, while the trend in suspensions

[7] We note, however, that the enrollment patterns in Provision 2 diverge between treatment and control schools. Nearly half of treatment school students in 2014 had exposure to some level of universal free breakfast or lunch supports via Provision 2 prior to the implementation of CEP, while just 13 percent of students in control schools did. The enormous drop off in Provision 2 participation among treatment schools by 2017 is mechanical – CEP schools must exit Provision 2 – while control schools saw slight growth in Provision 2 use, suggesting it remains an attractive alternative to CEP in certain cases.

among students in control schools levels off and even ticks upward in 2017. By 2017, despite having higher rates of poverty and attending schools with higher concentrations of poverty, students in schools that implemented CEP have lower rates of suspension compared to students in schools that declined to participate in CEP.

[FIGURE 2 AROUND HERE]

We formally test and graph pre-trends in Table 3 and Figure 3. We conduct this test by running a single regression of suspension on indicators for academic year, treatment cohort, and the interaction of the two. In Table 3, the first column of coefficients documents the trend in suspension rates for students in control group schools, while subsequent columns show the deviation from the control group trend for each treatment cohort. Specifically, the coefficients in bold offer our test of pre-trends: where they are small in magnitude and not statistically significant, we find no evidence of systematic divergence in suspension outcomes between treatment and control groups in the periods prior to treatment. We plot these coefficients in Figure 3, with each cohort-specific line weighted to evoke the number of schools in the cohort.

[TABLE 3 ABOUT HERE]

The coefficients on the cohort interactions in years prior to treatment are particularly close to zero for the 2015 cohort (as shown in the thick blue line in Figure 3), offering strong support of the parallel trends assumption for this cohort. The coefficients for the smaller 2016 and 2017 cohorts are less precise. For the 2016 cohort, coefficients are persistently negative, indicating that suspension rates started falling more rapidly in schools that implemented CEP in 2016 than they did in other CEP-eligible schools, even before these schools implemented the universal meals program. For the 2017 cohort, the opposite is true, and the generally positive coefficients indicate a flatter decline in suspension than the control schools. However, since

neither of these pre-trends divergences is significantly different from zero and since 77 percent of treatment schools are in the 2015 cohort, we see in Table 3 strong evidence in support of the parallel trends assumption that underlies the difference-in-differences analyses that we present below.

[FIGURE 3 ABOUT HERE]

Estimating the effects of CEP implementation

We next present the results of difference-in-differences models that answer our first two research questions, assessing the average effects of CEP implementation and estimating the extent to which these effects are sustained or grow in the years after treatment. Table 4 summarizes two sets of main effects: those from the disaggregated model, which allows treatment effects to vary by cohort and academic year, and those from the Model (1), which assumes a common treatment effect across cohorts for each period of treatment. We report results with and without covariates for both models. However, due to ongoing concerns in the literature about the interpretability and reliability of DiD-derived treatment effects in the presence of time-varying covariates, we do not forefront the version of the model that includes covariates (Caetano et al. 2022). We note, however, that including covariates consistently yields larger effect estimates.

[TABLE 4 AROUND HERE]

In the cohort-specific model (column 1), only the third treatment period (academic year 2016-17) for the 2015 cohort yields a statistically significant negative effect. This negative coefficient suggests that in its 3rd year of implementation CEP reduced suspension rates by approximately one percentage point (from a baseline rate of approximately 7.7 percent). The

coefficients for the first two treatment periods for both the 2015 and 2016 cohorts are negative, as expected, but imprecisely estimated.

To improve precision, we implement Model (1) by aggregating across cohorts and estimate a common effect of each treatment period. To justify this choice, though, we first test whether the first period effects are equal in the disaggregated model, and then whether the second period effects are. In both cases, we fail to reject the null hypothesis (see p-values at the bottom of column 1), so we proceed with the assumption of common period-specific effects across cohorts.

The results from Model (1) are shown in column 3. Although these results suggest that suspension rates decline by approximately 0.6 percentage points in the first and second years of CEP implementation, these coefficients are not statistically significantly different from zero. However, the coefficient for the third treatment period is statistically significant at the 1 percent level, and suggests that CEP reduces suspension rates by a full percentage point in its third year of implementation. Relative to the 7.7 percent baseline rate, our analysis suggests that universal meal provision reduces suspension in treated schools by approximately 8 percent in the first two years of CEP implementation and that this effect increases to approximately 13 percent in the third year of implementation. However, we cannot reject the null hypothesis that treatment effects are the same in each period (see p-value at bottom of column 3).

In summary, and consistent with our first hypothesis, we find that CEP implementation reduces suspensions in participating schools. Further, although our analysis suggests that CEP effects get larger as schools spend more time in the program, these coefficients are not significantly different from one another. Thus, our findings in support of our second hypothesis are qualified.

Evidence of effect heterogeneity

A unique contribution of our paper comes in our analysis of heterogeneity in treatment effects across subgroups of students, particularly those most likely to experience stigma. To study subgroups, we estimate Model (2) using OLS, and then report linear predictions of subgroup-specific treatment effects in Table 5. These estimates can be interpreted as the treatment effect on suspension of CEP for students in participating schools depending on their subgroup membership and the period of treatment.

[TABLE 5 AROUND HERE]

Although we do not have strong hypotheses concerning variation in CEP's effects across grade levels, our first subgroup analysis compares treatment effects for elementary and middle/high grades. There are two motivations for this analysis: first, middle/high schools tend to have much higher suspension rates than elementary schools, and, second, elementary schools are disproportionately likely to enroll in CEP.[8] These facts highlight a challenge for detecting the effect of CEP on suspension rates, since the schools theoretically capable of the largest reductions in suspensions rates are systematically less likely to enroll in CEP.

Panel A of Table 5 shows our grade-level-specific treatment effects. Both elementary and middle/high school students experience suspension reductions as a result of CEP. However, CEP's protective effects on suspensions are much more pronounced for students in middle and high school grades than students in elementary grades. To contextualize this finding, we note that among treatment schools in 2014, suspension rates were 14 percent in grades 6-12 compared to

[8] Elementary school students are disproportionately exposed to CEP in Oregon for two reasons. First, as Table 1 illustrates, a greater proportion of Oregon elementary schools are eligible for CEP. Second, take up of CEP is higher among elementary schools than middle and high schools. In our analytic sample of students enrolled in CEP-eligible schools, elementary grade students comprise 66 percent of students in treatment schools, in contrast to just 49 percent of students in control schools.

approximately 4 percent in grades K-5. Given that baseline difference, the third period effect of CEP is similar in proportional terms; both elementary and middle/high school students see suspension rate reductions of more than one-fifth.

Turning back to our central research questions, our third and fourth hypotheses suggests that reductions in suspensions following the introduction of universal free lunch should be concentrated among students most likely to experience lunch-related stigma: low-income students and students of color.

The analysis reported in Panel B of Table 5 focuses on the extent to which the effects of CEP vary with student economic background. We find that low-income students[9] experience large and statistically significant treatment effects. The provision of universal school meals under CEP reduces suspension rates for low-income students by a percentage point in the first year of program implementation. This effect is sustained in subsequent years and is statistically significant across all three treatment years. By contrast, we find no effect of CEP implementation on more affluent students. While the CEP effect coefficients for non-low-income students are negative across all three years, they are not statistically significant and are considerably smaller in magnitude than the coefficients for low-income students. In 2014, low-income students in CEP schools had suspension rates of 8.6 percent, compared to 5.6 percent for non-low-income students, so CEP may narrow the suspension rate gap by approximately one-third.

In panel C, we consider the extent to which the effects of CEP vary with student race and ethnicity. These estimates show that Hispanic students experience considerably larger treatment effects than students in all other racial/ethnic groups. These models suggest that CEP implementation reduces Hispanic students' suspension rates by 1.5 percentage points. The

[9] See Appendix A for the specific definition of this variable.

effects remain statistically significant and similar in magnitude in the second and third years of CEP implementation. While the effect estimates for White students in these models are also negative, they are much smaller than the estimated effects for Hispanic students and are not statistically significant. Effects for other racial subgroups are ambiguous – likely due to their small representation in Oregon public schools.

Thus, we find support for our third and fourth hypotheses: the protective effect of CEP on suspensions is largest for low-income students and students of color. These findings naturally raise questions concerning the overlap of these subgroups: are the Hispanic effects driven by low-income students? Are the low-income effects driven by Hispanic students? Do low-income White students experience protective effects? To answer these, we estimate race/ethnicity-by-income group-specific effects in Panel D. Several important nuances emerge. First, White low-income students experience sizable and significant benefits of CEP by the third treatment period, while Hispanic low-income students experience sustained benefits across treatment periods. Second, while non-low-income White students do not experience improvements in suspension, Hispanic students who are not low income still do.[10] This is consistent with lunch-related stigma operating along both socioeconomic and racial/ethnic dimensions.

Discussion

In this paper, we investigate how CEP, a federal program that allows schools to provide free meals to all students regardless of their family's socioeconomic background, affects participating schools' disciplinary climate.

[10] The other small racial groups are lumped together here in the "Other race" group for completeness, but we do not recommend interpreting those coefficients, as the subgroups included have very different baseline suspension rates

Our work brings together ideas from two separate strands of research on stigma in schools: First, we draw upon previous research that suggests the distribution of means-tested free and reduced-price school lunches raises the salience of students' family incomes, producing and reinforcing stigmatized identities associated with social class that follow students through their school days. Second, we draw upon scholarship linking stigma to racial disparities in exposure to exclusionary school discipline. This literature focuses primarily on the ways racialized stereotypes contribute to Black/White disciplinary gaps. We extend this "vicious cycle" theory to hypothesize that if universal free meal provision ameliorates socioeconomic stigma associated with free and reduced-price lunch participation, CEP implementation will (1) reduce the incidence of suspensions in participating schools, (2) producing larger protective effects over time, (3) particularly for students from low-income backgrounds and (4) students from minoritized racial and ethnic groups.

To test these hypotheses, we have constructed a unique student-by-year-level panel data resource that links more than 2 million annual records of K-12 students enrolled in Oregon public schools between 2009-10 and 2016-17 with rich administrative data records describing their families' household income and social safety net program participation. We use these data to estimate the effects of attending a school that has implemented CEP. Our models use a difference-in-differences design, in which we compare how student suspension rates change in the years following CEP implementation for participating schools versus those that were eligible to participate in the universal meals program but chose not to.

This estimation strategy hinges on the identifying assumption that, in the absence of CEP implementation, suspension rates in CEP-participating schools would have followed parallel trends to suspension rates in schools that did not participate in CEP. Our analyses suggest that

this identifying assumption holds. While suspension rates were falling rapidly across Oregon during the years leading up to CEP's launch, we find no evidence to suggest that the rate of change diverged between schools that implemented CEP and those that did not in the pre-treatment period.

Consistent with our first hypothesis, we find that CEP implementation has a measurable protective effect on suspension rates in participating schools. While our estimates of the immediate effects of CEP implementation are imprecise, we find that by the third year of implementation, CEP reduces suspension rates by a full percentage point, a statistically significant effect that equates to an approximately 13 percent reduction in suspensions. The trend toward increasing effects of CEP over time suggests that the implementation of CEP may initiate a longer term process of cultural change in participating schools, consistent with our second hypothesis. The CEP effects that we identify are somewhat smaller than the protective effect associated with rigorously evaluated large scale restorative justice implementation efforts (Augustine et al. 2018; Davidson et al. 2022). However, we do not see CEP as an alternative to these more targeted programs. Rather, we suspect that universal meal provision – a policy designed to improve student nutrition and reduce administrative burden – can contribute to a broader effort to reform school disciplinary climates.

Furthermore, we find that CEP implementation has particularly pronounced protective effects on suspensions for students from low-income families, defined as students whose families either participated in SNAP or reported total household income less than 1.85 times poverty in IRS tax filings. Our ability to estimate program effects on low-income students is possible because our unique linked data enables us to consistently identify low-income students

regardless of the practices of their schools. We see this as an important contribution to the literature on universal meal provision.

We further find that CEP has more pronounced protective effects for Hispanic students than White students. In summarizing these findings, we note that we are unable to estimate the effect of CEP implementation on Black students with precision. This is a major limitation, since anti-Black racism is a large and deeply-embedded source of inequality in school justice and substantial attention has been dedicated to narrowing persistent Black/white disciplinary disparities. Our findings cannot speak convincingly to whether or not CEP implementation can contribute to this effort. Future research should investigate whether these effects hold in settings with larger Black student populations.

Viewing universal meal provision as an intervention that mitigates a source of stigma in the school helps to explain how the policy's effects reach beyond student nutrition to promote equity at scale in the disciplinary arena. Traditional school meal programs create categories based on students' reported family income and then enact those categories daily as students claim their meals (Domina, Penner, & Penner 2017). But these categories are malleable. Universalizing school meals via CEP renders the distinction between students who receive free or reduced-price meals and other students meaningless by collapsing all students into the free meal category. Our analyses indicate that the effects of this adjustment in educational category ripple through school cultures to the particular benefit of groups of students who we might expect to be particularly at risk for income-based stigma.

Of course, CEP does not just manipulate the social categories that operate in schools; it also makes it easier for all students to claim free school meals. Since previous research indicates that lunch stigma is a primary deterrent to school meal consumption – for students who qualify

for free meals as well as for students who would have to pay for school meals – we do not see CEP's nutritional effects as theoretically distinct from its stigma-reducing effects. However, with appropriate data, it would be possible to empirically separate the ways CEP and other universal meal provision programs influence student outcomes via purely social mechanisms from the ways they influence outcomes via nutrition. While our data do not allow us to conduct this analysis, we hope that future scholars will. Similarly, our data do not allow us to distinguish CEP's effect on student behavior from its effects on teacher responses to student behavior. This is an important consideration for future work.

As more and more schools adopt the CEP and concerns about food insecurity mount, political pressure to extend the program and offer free meals to all students is growing. In the Spring of 2020, as schools across the US transitioned to remote-only instruction in an attempt to slow the spread of COVID-19, the USDA enacted a waiver that allowed schools to provide free meals to all children, regardless of income eligibility. In the pandemic's wake, policymakers at the state and federal level are considering a wide range of policies to further expand access to free school meals. It is not clear that our findings regarding the effect of a program that is available only to schools with relatively high concentrations of poor students generalize to a universal free meal program that includes all public schools. However, we believe our findings provide a strong indication of the harms associated with lunch stigma and the potential to create more equitable schools by addressing those harms.

Appendix A: Variable Construction

Table 1 provides an overview of our data and key variables. Student-level variables include information on our key outcome – suspension – as well as demographic characteristics (from ODE data), school mobility rates (ODE data), SNAP enrollment and benefits (Oregon SNAP data), and family income (IRS data). School characteristics are derived primarily from ODE data, but school median adjusted gross income is measured from IRS records. School use of Provision 2 – which indicates participation in a different NSLP program that facilitated universal breakfast and lunch provision in certain cases prior to CEP – is identified from public records.

While the variables from ODE are standard in K-12 administrative data, the variables derived from SNAP and IRS records require more detailed explanation.

Enrolled in SNAP in March. This indicates students whose households received nonzero SNAP benefits in March of the relevant school year. We use March SNAP receipt, because SNAP enrollments as of April 1 of each year are used to determine school eligibility for CEP, and March receipt was our best proxy for that point-in-time measure.

Total annual family SNAP benefits. This is the total SNAP benefits received by the student's household from August through July (corresponding with the relevant school year).

Adjusted gross income (back-filled). This is the adjusted gross income reported on the tax return (IRS Form 1040) on which the student is claimed as a dependent. Tax records reflect calendar years, so we link to the tax record corresponding to fall of each school year (e.g., student records for the 2014-15 school year are linked to 2014 tax records). Since some households do not file taxes each year, we take the adjusted gross income from the most recent tax record on which the student appears, up to four years prior to the reference year. To ensure comparability, we adjust

the prior year records for inflation using the Consumer Price Index, so that any backfilled income value is reported in reference year dollars.

Low income. This is a composite indicator that leverages both Oregon SNAP and IRS records. Specifically, students are designated as low income if (a) they are enrolled in SNAP at any point during the school year[11], or (b) they are not enrolled in SNAP but their family's income-to-poverty ratio falls below 1.85 times the poverty line. To calculate the family income-to-poverty ratio, we assign a poverty line based on family size proxied from the student's most recent tax return record, and divide the adjusted gross income (back-filled) value by the poverty line.

[11] Typical SNAP guidelines require households have income less than 1.3 times the federal poverty line, but here, since we include households that may have only been SNAP-eligible for part of the year, we designate as low-income some households with income above 1.3 times the poverty line, and, potentially, above 1.85 times the poverty line. However, our analyses suggest that the vast majority of these families have annual income below 1.85 times poverty.

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Table 1: Population and sample summary

	All Oregon schools		CEP-eligible schools	
	Mean	Std. dev.	Mean	Std. dev.
A. Student characteristics				
Ever suspended	0.06	0.25	0.08	0.26
Suspension days	0.21	1.28	0.24	1.39
Female	0.49	0.50	0.49	0.50
American Indian/Alaskan Native	0.02	0.13	0.02	0.14
Asian/Pacific Islander	0.05	0.21	0.04	0.20
Black	0.02	0.16	0.03	0.17
Hispanic	0.22	0.41	0.26	0.44
White	0.65	0.48	0.60	0.49
Multiracial	0.05	0.22	0.05	0.21
Moves schools between years (nonstructural)	0.09	0.28	0.09	0.29
Moves schools during year	0.05	0.21	0.05	0.22
Language besides English spoken at home	0.22	0.41	0.27	0.44
Enrolled in SNAP in March	0.29	0.46	0.41	0.49
Total annual family SNAP benefits	1,491	2,573	2,093	2,870
Adjusted gross income (backfilled)	71,760	168,900	49,230	67,910
Low income	0.35	0.48	0.47	0.50
B. School characteristics				
Provision 2	0.10	0.30	0.19	0.39
Avg. school enrollment	721	564	634	484
Avg. district enrollment	15,270	15,320	13,190	14,270
Avg. school median AGI	54,290	27,860	37,930	9,237
Avg. proportion grades K-5	0.46	0.50	0.56	0.50
Avg. proportion grades 6-8	0.23	0.42	0.21	0.41
Avg. proportion grades 9-12	0.31	0.46	0.23	0.42
C. Sample description				
No PIK	0.04	0.19	0	
No low income value	0.06	0.24	0	
N(observations) All	4,483,000			
N(students) All	1,013,000			
N(observations) with PIK	4,312,000			
N(students) with PIK	956,000			
N(observations) with PIK & Low Income value	4,219,000		2,104,000	
N(students) with PIK & Low Income value	945,000		567,000	

Note: "Avg." school characteristics are measured at school level but averaged at student level, resulting in a weighted average of school characteristics in each cell. Estimates and counts are rounded according to Census disclosure guidelines.
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Table 2: Trends in covariates for treatment and control schools

Table 2: Trends in covariates for treatment and control schools

	Ever in CEP				Never in CEP			
	2010	2014	Change (2014-2010)	2017	2010	2014	Change (2014-2010)	2017
A. Student characteristics								
Female	0.486	0.487	0.001	0.490	0.487	0.487	0.001	0.485
American Indian/Alaskan Native	0.031	0.025	-0.006	0.023	0.020	0.017	-0.004	0.014
Asian/Pacific Islander	0.043	0.042	-0.001	0.040	0.038	0.039	0.001	0.038
Black	0.041	0.038	-0.003	0.038	0.026	0.024	-0.003	0.023
Hispanic	0.298	0.335	0.038	0.344	0.173	0.210	0.036	0.222
White	0.557	0.514	-0.043	0.505	0.708	0.657	-0.050	0.644
Multiracial	0.024	0.046	0.022	0.051	0.029	0.054	0.025	0.059
Moves schools between years (nonstructural)		0.117		0.106		0.102		0.096
Moves schools during year	0.060	0.056	-0.004	0.047	0.048	0.047	-0.001	0.041
Language besides English spoken at home	0.326	0.339	0.013	0.340	0.192	0.203	0.011	0.220
Enrolled in SNAP in March	0.448	0.490	0.042	0.416	0.333	0.369	0.036	0.312
Total annual family SNAP benefits	2,405	2,539	134	2,070	1,730	1,878	148	1,512
Adjusted gross income (backfilled)	40,510	42,680	2,170	49,410	50,010	52,930	2,920	60,830
Low income	0.504	0.561	0.058	0.492	0.385	0.435	0.050	0.379
B. School characteristics								
Provision 2	0.398	0.493	0.096	0.013	0.114	0.125	0.011	0.137
Avg. school enrollment	522	534	12	539	720	715	-4	713
Avg. district enrollment	14,510	14,970	460	15,260	11,470	11,820	350	12,140
Avg. school median AGI	31,770	32,720	950	37,650	39,580	40,660	1,080	46,480
Number of students	116,000	118,000	2,000	117,000	145,000	147,000	2,000	145,000
Number of schools	300	300	0	300	350	350	0	350

Note: "Avg." variables are measured at school level but averaged at student level, resulting in a weighted average of school characteristics in each cell; counts are rounded according to Census disclosure guidelines.
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Table 3: OLS, Suspension rate pre-trends

		Divergence from control group trend for…		
Academic year	Trend for control group	2015 cohort	2016 cohort	2017 cohort
2011	0.0000593	**-0.00385**	**-0.0103***	**-0.00128**
	(0.00371)	**(0.00432)**	**(0.00480)**	**(0.00497)**
2012	-0.00596*	**0.000564**	**-0.0149**	**0.00828**
	(0.00275)	**(0.00484)**	**(0.0127)**	**(0.0106)**
2013	-0.0132***	**-0.00238**	**-0.0156**	**0.0123**
	(0.00380)	**(0.00529)**	**(0.0141)**	**(0.00794)**
2014	-0.0203***	**-0.00147**	**-0.0197**	**0.0132**
	(0.00344)	**(0.00737)**	**(0.0138)**	**(0.0103)**
2015	-0.0229***	-0.00583	**-0.0160**	**0.0102**
	(0.00387)	(0.00618)	**(0.0196)**	**(0.00623)**
2016	-0.0239***	-0.00509	-0.0250	**0.0131**
	(0.00367)	(0.00637)	(0.0185)	**(0.0126)**
2017	-0.0199***	-0.0116*	-0.0292	0.00810
	(0.00420)	(0.00523)	(0.0174)	(0.0141)
Constant	0.0913***			
	(0.00259)			
School FEs	Yes			
Covariates	No			
Observations	2104000			
r2	0.0522			

Note: this table displays the full set of coefficients from a single regression. **Pre-treatment academic years in bold**. The "Trend for control group" column displays coefficients on academic year indicators; the "2015 cohort" column displays coefficients on an indicator for membership in the 2015 treatment cohort interacted with academic year indicators. The reference (omitted) year is 2010. Cohort indicators are subsumed by school fixed effects. Standard errors in parentheses. * p<0.05; ** p<0.01; *** p<0.001.
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Table 4: OLS, Suspension rate treatment effects

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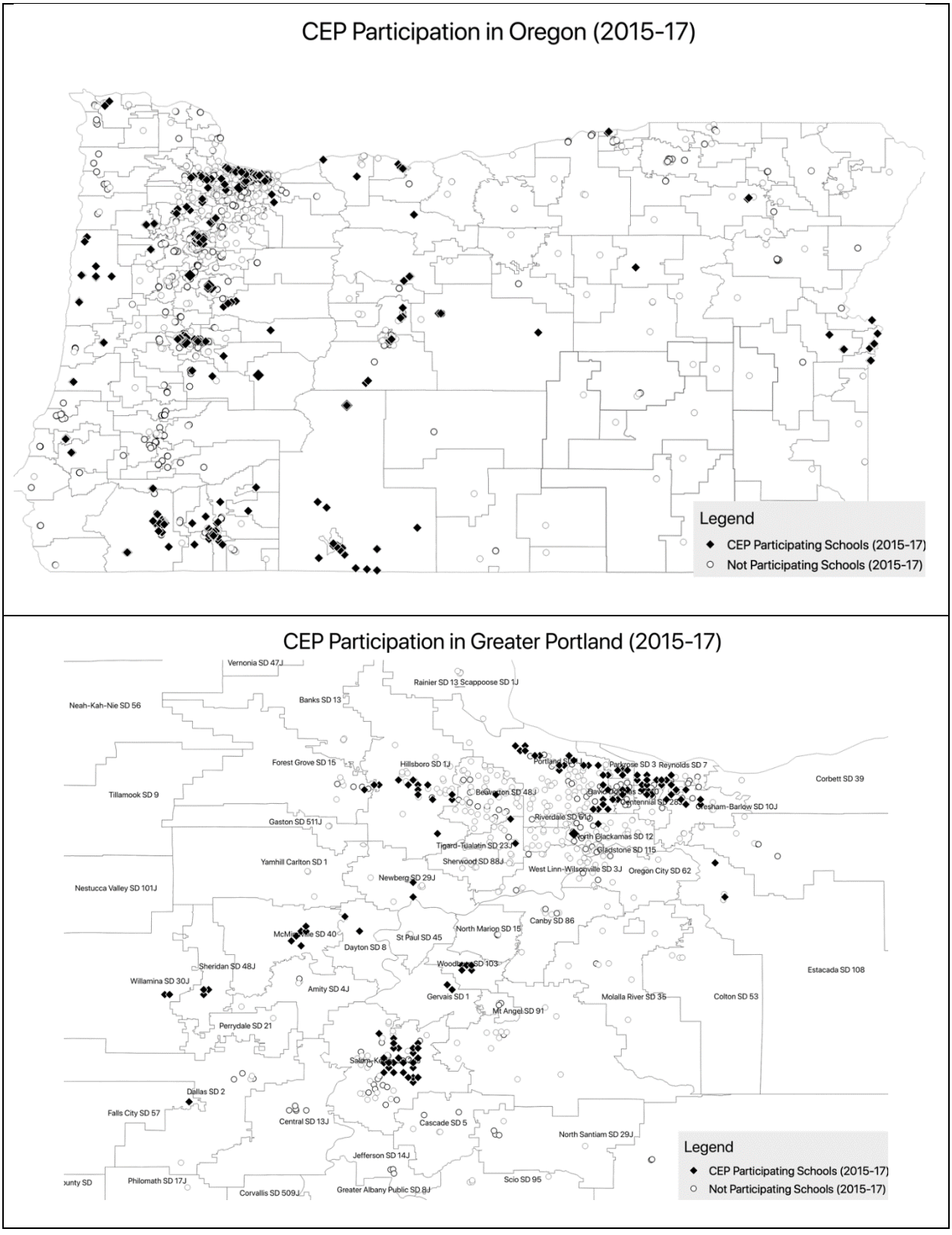
	(1)	(2)	(3)	(4)
Cohort 2015 in year 2015	-0.00398	-0.00727		
	(0.00430)	(0.00507)		
Cohort 2015 in year 2016	-0.00372	-0.00669		
	(0.00373)	(0.00453)		
Cohort 2015 in year 2017	-0.0101***	-0.0130**		
	(0.00296)	(0.00441)		
Cohort 2016 in year 2016	-0.0124	-0.0125		
	(0.00895)	(0.00890)		
Cohort 2016 in year 2017	-0.0164	-0.0163		
	(0.00920)	(0.00942)		
Cohort 2017 in year 2017	0.000292	0.000981		
	(0.0100)	(0.0100)		
First period of treatment			-0.00605	-0.00872
			(0.00375)	(0.00500)
Second period of treatment			-0.00622	-0.00900*
			(0.00346)	(0.00447)
Third period of treatment			-0.00977**	-0.0132**
			(0.00303)	(0.00443)
Constant	0.0765***	0.330**	0.0766***	0.327**
	(0.000458)	(0.111)	(0.000455)	(0.113)
School and academic year fixed effects	Yes	Yes	Yes	Yes
Covariates	No	Yes	No	Yes
Observations	2104000	2058000	2104000	2058000
r2	0.0522	0.0859	0.0522	0.0859
P-value, test: first period effects are equal	0.641	0.608		
P-value, test: second period effects are equal	0.213	0.320		
P-value, test: effects are equal across periods			0.427	0.338

Note: Standard errors in parentheses; * p<0.05; ** p<0.01; *** p<0.001.
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Table 5: Linear predictions, Heterogeneous treatment effects

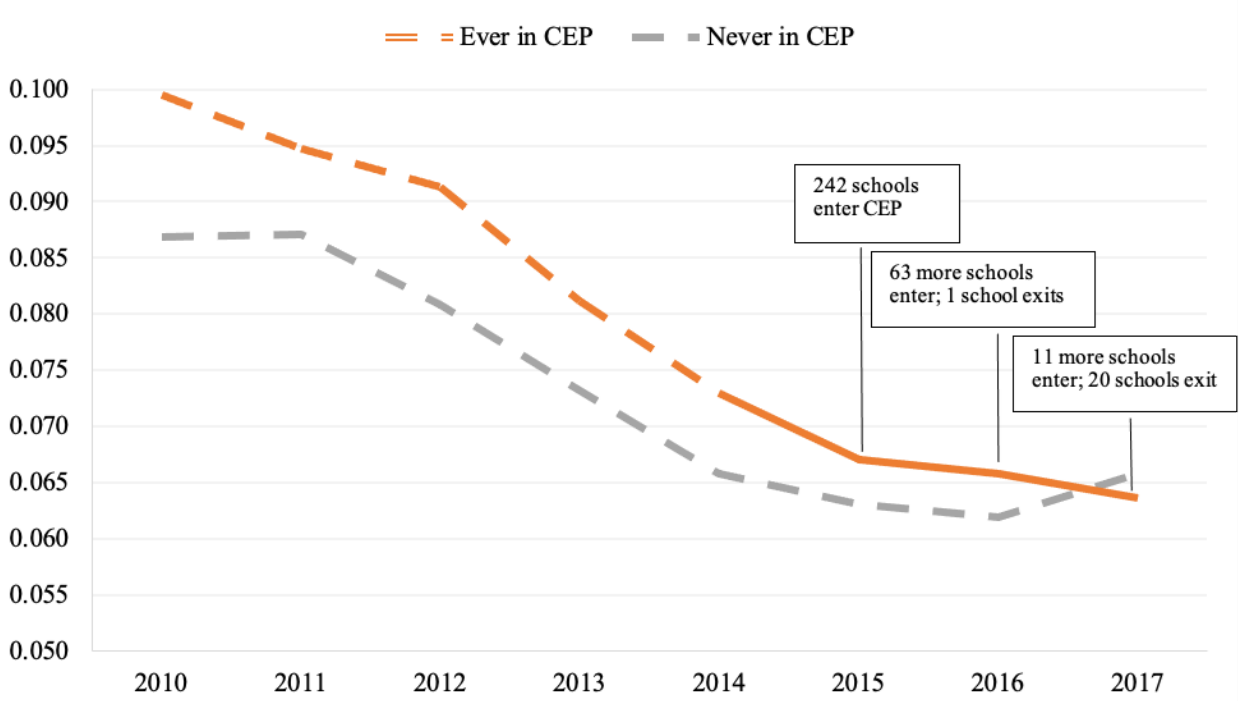
	Period 1	Period 2	Period 3
A. Grade			
Grades K-5	-0.00513	-0.00402	-0.00913**
	(0.00277)	(0.00278)	(0.00299)
Grades 6-12	-0.0174*	-0.0196**	-0.0237***
	(0.00753)	(0.00733)	(0.00720)
B. Student income status			
Not low income	-0.00107	-0.00147	-0.00424
	(0.00411)	(0.00378)	(0.00378)
Low income	-0.0101*	-0.0100*	-0.0142***
	(0.00433)	(0.00462)	(0.00398)
C. Student race			
American Indian/Alaskan Native	0.00825	0.0240	0.00955
	(0.0192)	(0.0277)	(0.0243)
Asian/Pacific Islander	-0.00162	-0.000249	0.0133
	(0.00651)	(0.00638)	(0.00949)
Black	-0.00200	0.00373	-0.0102
	(0.00747)	(0.0105)	(0.0115)
Hispanic	-0.0151*	-0.0123**	-0.0143***
	(0.00591)	(0.00451)	(0.00318)
Multiracial	-0.00262	0.00170	0.00327
	(0.00540)	(0.00587)	(0.00600)
White	-0.0000340	-0.00256	-0.00699
	(0.00305)	(0.00393)	(0.00384)
D. Student income and race			
White, Not low income	0.00248	0.000633	-0.00365
	(0.00403)	(0.00440)	(0.00472)
White, Low income	-0.00535	-0.00862	-0.0132**
	(0.00341)	(0.00549)	(0.00476)
Hispanic, Not low income	-0.00989	-0.00915	-0.00943*
	(0.00541)	(0.00500)	(0.00468)
Hispanic, Low income	-0.0162*	-0.0123**	-0.0160***
	(0.00634)	(0.00461)	(0.00345)
Other race, Not low income	-0.00124	0.00563	0.00448
	(0.00664)	(0.00597)	(0.00557)
Other race, Low income	0.000520	0.00432	0.00345
	(0.00771)	(0.00892)	(0.00729)

Note: Estimates are linear predictions derived from regressions fully-interacted with the characteristic(s) of interest. Each panel reports results from a different regression. Standard errors in parentheses. * p<0.05; ** p<0.01; *** p<0.001. Approved under CBDRB-FY22-CES014-012



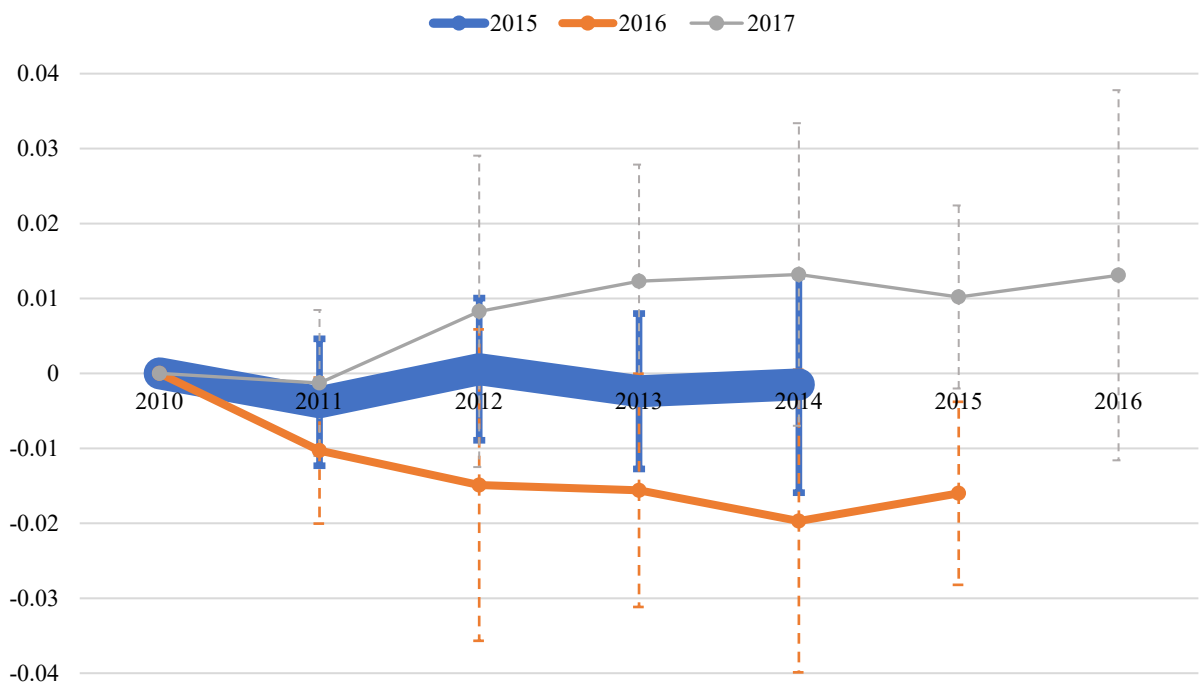
Source: Oregon Department of Education.

Figure 1: Map of schools by CEP participation



Note: Approved under CBDRB-FY22-CES014-012

Figure 2: Trends for suspension rates



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Figure 3: OLS coefficients, Suspension rate pre-trends